UNIT 7 · STANDARD 6.NOS.7

Reflect Points Across Axes

Lesson 7-9

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes Level 2 Standard

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can reflect points across the x-axis and y-axis on the coordinate plane.

Language Objective: I can explain my work using the words reflection, x-axis, y-axis, and symmetry.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Reflection | x-axis | y-axis | Symmetry | Integer | Perpendicular

☝️ *Tap any word to see its meaning and a picture.*

1. A flip of a point or shape over an axis to make a mirror image is a _____.

2. The horizontal number line on the coordinate plane is the _____.

3. The vertical number line on the coordinate plane is the _____.

4. When two halves are mirror images of each other, the figure has _____.

5. A whole number or its opposite, with no fraction part, is an _____.

6 Two lines that meet at a 90° square corner are _____ .

Write About the Math

How do reflections help the game designer create a balanced, symmetric level?

¿Cómo ayudan los reflejos a la diseñadora del juego a crear un nivel equilibrado y simétrico?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Reflection** — *Reflexión*
- ☐ **x-axis** — *Eje x*
- ☐ **y-axis** — *Eje y*
- ☐ **Symmetry** — *Simetría*
- ☐ **Integer** — *Número entero*

Model: *Reflecting over the y-axis changes the ____ coordinate because ____.* — Student states reflecting over the y-coordinate flips the sign of the x-coordinate while the y-coordinate stays the same, so (3,2) becomes (-3,2).

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The enemy reflected over the y-axis is at ____ because ____. The enemy reflected over the x-axis is at ____ because ____. Reflections help game designers because ____.**
- **My answer is ____ because ____.**
  Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **My evidence is ____, so ____.**
  Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.